XenDirekt — Product Requirements Document (MVP v1.0)

Oct 18, 2026 ·

1. Overview

XenDirekt is a mobile app that lets an international visitor pay any Rwandan MTN or Airtel MoMo account with the Visa or Mastercard card they already carry. The merchant receives Rwandan francs (RWF) in their existing MoMo account and needs nothing new: no app, no sticker, no POS, no sign-up.

What XenDirekt is: a pass-through bridge between international card networks and Rwandan mobile money. **What it is not:** a wallet, a bank or a stored-value product. XenDirekt never holds a customer balance; every payment moves from the card to the MoMo account in one transaction, through licensed partners.

Item	Value
Product	XenDirekt mobile app (sender side only) + back-end payment service + internal admin console
Release covered	MVP v1.0 (pilot in Kigali)
Product owner & PM	XenDirekt team
Audience	Software developers, designer, QA, payment-partner technical contacts
Tagline	You don't have to worry whether your payment means work or not before you enter any shop.

How to read this PRD. Requirements carry IDs (for example FR-PAY-03) so tickets and tests can reference them. Priority uses MoSCoW: **Must** = needed for the pilot, **Should** = wanted for the pilot if time allows, **Could** = later. Anything marked *Open question* needs a decision from the PM before it is built.

2. Problem, goals and success metrics

Problem. First-time, short-stay international leisure tourists in Kigali cannot pay small shops and market stalls with their bank card, because those businesses have no POS and accept only cash or MoMo. Tourists lose time looking for an ATM, a FOREX bureau or another shop, pay ATM fees, and sometimes abandon the purchase. This was observed in Gatenga (Kigali) and confirmed in 20 tourist interviews.

Product goals (MVP)

1. A tourist can pay any MTN or Airtel MoMo account in Rwanda from an international card in under 60 seconds, without help.
2. The tourist sees the merchant's registered name before paying, so they trust they are paying the right person.

3. Both sides get a clear confirmation (in-app + SMS) the moment the money lands.
4. XenDirekt never holds customer funds and operates under a licensed partner.
5. Fraud and chargeback losses stay below an agreed limit (see section 8).

Non-goals (MVP)

- No merchant app, merchant dashboard, QR stickers or merchant onboarding.
- No stored balance, wallet top-up or cash-out for tourists.
- No USSD channel (the tourist has no local SIM).
- No payments outside Rwanda, and no Rwandan-to-Rwandan payments.
- No loyalty, rewards or marketplace features.

Success metrics

Metric	Definition	Pilot target
Payment completion rate	Payments that reach SUCCESS ÷ payments where the tourist tapped Pay	≥ 60% unaided (go); < 40% means rework
Time to pay	From opening the Pay screen to the success screen, median	≤ 60 s
Onboarding completion	Sign-ups that add a working card ÷ app installs	≥ 70%
Repeat use	Tourists who make a 2nd payment within their trip	≥ 50%
Settlement time	Card authorised → MoMo credited, 95th percentile	≤ 30 s
Fraud / chargeback rate	Disputed or fraudulent value ÷ total processed value	< 0.5%

The pilot thresholds come from the Week 5 success metrics (go if at least 6 of 10 attempted payments complete without help). Targets after the pilot are set by the PM from pilot data.

3. Users and the end-to-end journey

User	Who they are	Uses XenDirekt?	What they need
Tourist (sender)	First-time, short-stay international visitor in Kigali with a foreign Visa/Mastercard and a smartphone with data (roaming, eSIM or Wi-Fi)	Yes, the only app user	Pay a small shop fast, trust the payee, get proof of payment
Merchant (receiver)	Small shop, craft shop, market stall, moto or taxi driver with an MTN or Airtel MoMo account (personal number or MoMo Pay merchant code)	No	Receive RWF in their usual MoMo account and see an SMS they trust
XenDirekt operations	Internal staff	Admin	Monitor payments, handle support tickets, review

User	Who they are	Uses XenDirekt?	What they need
agent		console only	flagged payments, run reconciliation

Persona quote (research): "It wasn't the price that was the problem. It was figuring out how to actually pay." — Oliver, Canada.

Happy path, as the tourist experiences it

1. Downloads XenDirekt, signs up with email or phone, verifies it with a one-time code, and adds a card (once per trip).
2. At a shop, the merchant says the price and gives their MoMo number or MoMo Pay code.
3. The tourist opens **Pay**, types the number or code and the amount in RWF; the network (MTN or Airtel) is detected automatically.
4. On the same screen, the app looks up the account instantly and shows the merchant's registered name, the amount in RWF, the equivalent in the card's currency, and the XenDirekt fee.
5. The tourist checks the name with the merchant and taps **Pay**; the card bank may ask for 3-D Secure (OTP or bank-app approval).
6. The card is charged, the RWF amount is credited to the merchant's MoMo account, and both receive an SMS. The tourist sees a success screen and a receipt.

The merchant never touches XenDirekt: their only experience is the normal MoMo credit SMS from their operator plus a XenDirekt SMS naming the payer and reference.

4. Scope

The MVP is one sender-side mobile app, one payment back end and one internal admin console, launched as a pilot in Kigali.

Area	In scope (MVP)	Out of scope (later or never)
Platforms	Android and iOS app (one cross-platform codebase)	Web checkout (phase 2), merchant app (never planned)
Payers	Holders of Visa and Mastercard debit/credit cards issued outside Rwanda	Rwandan-issued cards, Amex/UnionPay (later), bank transfer, Apple Pay/Google Pay (phase 2)
Payees	Any active MTN MoMo or Airtel Money account in Rwanda: phone number or MoMo Pay merchant code	Bank accounts, payees outside Rwanda
Currency	Tourist charged in card currency (or USD), merchant paid in RWF	Multi-currency payout
Money flow	Card → partner → MoMo in one transaction	Stored balance, top-up, tourist cash-

Area	In scope (MVP)	Out of scope (later or never)
		out
Support	In-app help, FAQ, contact form, payment reference on every receipt	Live chat, phone line
Refunds	Manual refund of a card charge by ops when the MoMo credit failed	Merchant-initiated refunds, in-app dispute flow
Languages	English	Kinyarwanda, Swahili, German, Chinese (later)
Geography	Rwanda (pilot in Kigali)	Other MoMo countries (Uganda, Ghana, etc.)

Partner model. XenDirekt does not connect to card schemes or mobile operators directly in the MVP. It integrates with one licensed partner (an aggregator or PSP) that offers both card acceptance and MoMo payouts in Rwanda. Candidates are listed in section 7.

5. Functional requirements

Eight modules make up the MVP; every requirement below is a Must unless marked otherwise.

5.0 The three-tap rule (applies to every requirement below)

Every user action in the app completes in **3 taps or fewer**, counted from the open, unlocked app to the success state; a payment to a new merchant takes 1 tap.

How taps are counted. A tap is any press on a button, chip, toggle or list item. Typing a number, an amount or a PIN is not a tap, but is kept to the minimum. Two things outside XenDirekt's control are not counted: the card bank's 3-D Secure screen and first-time phone permission prompts (camera, notifications).

Design rules the developer must follow

1. **The app opens on the Pay screen.** No home menu to pass through; three bottom tabs only: Pay, History, Profile.
2. **Nothing nested deeper than one level.** Every setting is one item on the Profile screen; no sub-menus.
3. **Auto-advance.** When an input is complete (10-digit MoMo number, 6-digit code, PIN), the app moves on by itself. No Next, Search, Submit or Save buttons.
4. **Autofill everything possible.** One-time codes read automatically, name and email from Apple/Google sign-in, country from the device, card details by camera scan or phone autofill, network (MTN/Airtel) from the number prefix.
5. **Smart defaults.** Default card preselected, recent payees shown as one-tap chips, last amount offered on Pay again.

6. **No confirmation screens or "Are you sure?" pop-ups.** The summary (name, fee, total) appears live on the Pay screen; risky actions use Undo for 5 seconds instead. Only Delete account uses hold-to-confirm.
7. **Biometrics replace PIN typing.** Face ID or fingerprint starts automatically when Pay is tapped; PIN only as fallback.
8. **Deep links.** A push notification or SMS link opens the exact payment or receipt.

Tap map (target taps per action)

Action	Taps	How
Pay a new merchant	1	App opens on Pay → type number (name appears) and amount → **Pay** → Face ID
Pay a recent merchant	2	Payee chip → type amount → **Pay**
Pay again, same merchant and amount	2	**Pay again** on a recent payment → **Pay**
Pay with a different card	3	Card chip → choose card → **Pay**
Retry a failed payment	1	**Try again** on the failure screen (same quote reused if still valid)
Sign up	3	**Continue with Apple/Google** → code autofilled, details prefilled → **Continue** → type PIN → **Enable Face ID**
Link a bank card	2	**Add card** → scan card with camera or autofill → **Save card** → bank 3-D Secure
View a receipt	1	Tap the payment on Pay or History
Share a receipt	2	Payment → **Share**
Open a payment from a notification	1	Tap the notification
Report a problem	3	Payment → **Report a problem** → reason chip (sent with the reference attached)
Change PIN	2	Profile → **Change PIN** → Face ID confirms → type new PIN twice (saves itself)
Turn biometrics on or off	2	Profile → toggle
Set default card	3	Profile → card → **Make default**
Remove a card	3	Profile → card → **Remove** (Undo for 5 s)
Verify identity when asked	2	**Verify now** → passport and selfie auto-captured → **Submit**

Action	Taps	How
Change language	3	Profile → Language → pick
Read a help article	3	Profile → Help → question
Log out	2	Profile → **Log out**
Delete account	3	Profile → **Delete account** → hold to confirm

Admin console. The same rule applies where possible: search by typing, then payment → **Refund** → confirm (3), and payment → **Block** (2).

Acceptance. QA tests every row of the tap map on Android and iOS; a flow that needs more taps than listed is a bug. The app logs a `taps_to_complete` value per flow so the pilot can confirm it.

5.1 Onboarding and account

ID	Requirement	Priority
FR-ONB-01	Splash screen with the XenDirekt logo (under 2 s), then one welcome screen with the sign-in buttons; no intro carousel	Must
FR-ONB-02	Sign up with Apple, Google, email or international phone number (E.164); the one-time code (6 digits, 5-minute expiry, max 5 attempts) is read automatically (Android SMS Retriever, iOS code autofill, email magic link) and submits itself	Must
FR-ONB-03	Collect full name, country of residence and date of birth, prefilled from Apple/Google sign-in and the device region; Terms and Privacy are accepted by the same Continue tap, with a timestamped record	Must
FR-ONB-04	Set a 4–6 digit app PIN (saves itself on the last digit) and turn on Face ID / fingerprint in the same step; after that, biometrics replace PIN typing everywhere	Must
FR-ONB-05	Language choice: English, French; default from device language	Must
FR-ONB-06	Identity check (passport photo + selfie liveness) through the partner's KYC service, triggered only when limits in section 8 are exceeded	Must
FR-ONB-07	Delete account from settings; data kept only as long as the law requires	Must

5.2 Card management

ID	Requirement	Priority

ID	Requirement	Priority
FR-CRD-01	Add a Visa or Mastercard through the partner's tokenisation SDK or hosted fields; card numbers never touch XenDirekt servers; offer camera card scanning and Apple Pay / Google Pay card autofill where the partner SDK supports it	Must
FR-CRD-02	Verify the card with a 3-D Secure authentication and a zero-amount or small reversible authorisation	Must
FR-CRD-03	Reject cards issued in Rwanda, prepaid gift cards and blocked BIN ranges (configurable list)	Must
FR-CRD-04	Show saved cards as brand + last 4 digits + expiry; set a default; remove a card	Must
FR-CRD-05	Up to 3 saved cards per user	Must

5.3 Payee lookup

ID	Requirement	Priority
FR-PAY-01	Pay screen offers two input types: MoMo phone number (+250 7X XXX XXXX) or MoMo Pay merchant code	Must
FR-PAY-02	Detect the network (MTN or Airtel) from the number prefix; let the user override	Must
FR-PAY-03	Call the partner's account-name lookup automatically as soon as the number is complete (no Search button) and show the registered account name before any charge	Must
FR-PAY-04	If lookup fails or the account is inactive, block payment and show a clear message ("We couldn't find this MoMo account. Check the number with the seller.")	Must
FR-PAY-05	Recent payees list (name + masked number) for one-tap repeat payments	Should
FR-PAY-06	Paste a number from the clipboard and strip spaces and the country code automatically	Should

5.4 Amount, quote and confirmation

ID	Requirement	Priority
FR-QTE-01	User enters the amount in RWF (whole francs); min and max amounts are server-configured	Must

ID	Requirement	Priority
FR-QTE-02	Server returns a quote: RWF amount the merchant receives, XenDirekt fee, FX rate, total charged in card currency; valid for 5 minutes	Must
FR-QTE-03	No separate confirmation screen: the Pay screen shows a live summary (merchant name, masked number, RWF amount, fee, total in card currency, card used) as soon as number and amount are entered	Must
FR-QTE-04	Optional note (max 60 characters) included in the merchant SMS, e.g. "2 baskets"	Should
FR-QTE-05	One tap on Pay starts biometric approval automatically (PIN only as fallback); the Pay button is disabled while a request is in flight (no double taps)	Must

5.5 Payment execution

ID	Requirement	Priority
FR-EXE-01	Every payment carries a unique idempotency key generated by the app; resending the same key never charges twice	Must
FR-EXE-02	Authorise the card (with 3-D Secure when the issuer asks), then request the MoMo payout, then capture the card only after payout success	Must
FR-EXE-03	If the payout fails, void or refund the card authorisation automatically and tell the user	Must
FR-EXE-04	Show live status (Processing → Sent → Received) and keep polling or listening to webhooks until a final state	Must
FR-EXE-05	A payment with no final state after 2 minutes moves to PENDING_REVIEW and is visible to ops	Must

5.6 Confirmations and receipts

ID	Requirement	Priority
FR-RCP-01	SMS to the merchant: "You received RWF 12,000 from [Tourist first name] via XenDirekt. Ref XD-7F3K9Q." (EN, with Kinyarwanda version)	Must
FR-RCP-02	SMS (or push + email if SMS can't reach a foreign number) to the tourist with amount, merchant name and reference	Must

ID	Requirement	Priority
FR-RCP-03	In-app receipt: reference, date/time (Kigali time), merchant name, RWF amount, fee, FX rate, card currency total, card last 4, status	Must
FR-RCP-04	Share or download the receipt as PDF/image	Should

5.7 History and support

ID	Requirement	Priority
FR-HIS-01	Payment history list, newest first, with status chips; filter by status	Must
FR-HIS-02	Help centre (FAQ) + "Report a problem" on any payment, which creates a ticket with the payment reference attached	Must
FR-HIS-03	Push notifications for payment success, failure and refund	Must

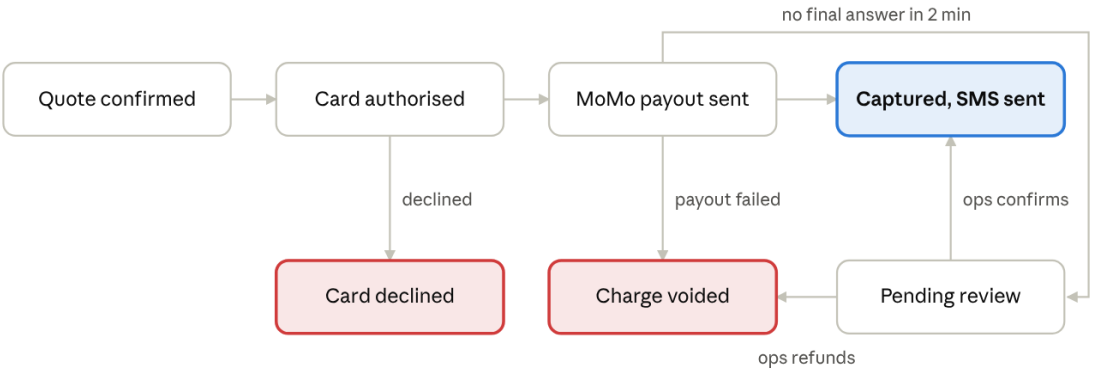
5.8 Admin console (internal web app)

ID	Requirement	Priority
FR-ADM-01	Login with SSO or email + 2FA; roles: Viewer, Support agent, Risk analyst, Admin	Must
FR-ADM-02	Search payments by reference, phone, email, card last 4, MoMo number, date; view the full timeline of states and partner responses	Must
FR-ADM-03	Review queue for PENDING_REVIEW and fraud-flagged payments; approve, refund or block	Must
FR-ADM-04	Block or unblock a user, a card fingerprint or a MoMo payee	Must
FR-ADM-05	Configure fees, limits, min/max amounts and blocked BINs without a release	Must
FR-ADM-06	Daily reconciliation report: XenDirekt records vs partner settlement file, with mismatches listed	Must
FR-ADM-07	Dashboard: volume, count, completion rate, median time to pay, failures by reason	Should
FR-ADM-08	Every admin action is written to an audit log (who, what, when, before/after)	Must

6. Payment flow, states and errors

The card is only authorised (held) first and captured after the MoMo credit is confirmed, so a tourist is never charged for money the merchant did not receive.

A card is captured only after the MoMo payout succeeds



payment states · happy path and 3 failure paths

A declined card ends the payment; a failed payout voids the hold automatically; a payment with no answer goes to ops, who either confirm it or refund it.

Transaction states (stored on every payment)

State	Meaning	Final?
QUOTED	Payee looked up, amount and fee locked for 5 minutes	No
AUTH_PENDING	Waiting for the card issuer / 3-D Secure	No
AUTH_FAILED	Card declined, 3-D Secure failed or timed out	Yes
AUTHORISED	Funds held on the card	No
PAYOUT_PENDING	MoMo credit requested from the partner	No
PAYOUT_FAILED	Partner or operator rejected the credit	No (goes to VOIDED)
VOIDED	Card hold released, user notified	Yes
PENDING_REVIEW	No final answer within 2 minutes; ops must resolve	No
COMPLETED	MoMo credited, card captured, SMS sent to both sides	Yes
REFUNDED	Card charge returned by ops after a completed or reviewed payment	Yes

Error handling (what the user sees)

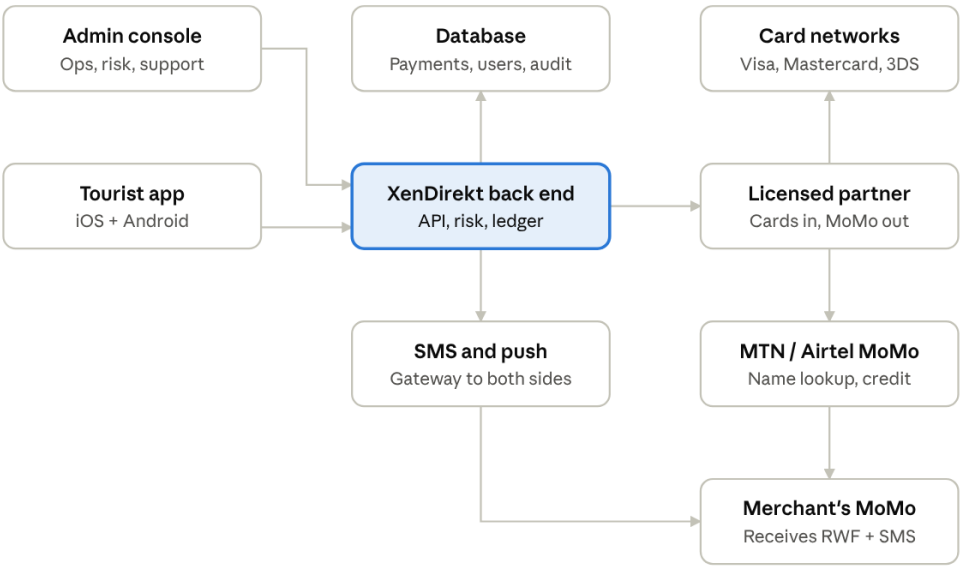
Situation	App message	System action

Situation	App message	System action
Number not found or inactive	We couldn't find this MoMo account. Check the number with the seller.	Block Pay; log lookup failure
Quote expired (> 5 min)	The price was updated. Please confirm the new total.	Fetch a new quote
Card declined	Your bank declined this payment. Try another card or contact your bank.	AUTH_FAILED; show decline reason if the partner gives one
3-D Secure abandoned	Payment not completed. You were not charged.	AUTH_FAILED after timeout
Payout failed	The seller's MoMo couldn't receive this payment. You have not been charged.	Void the hold; notify the user
Network lost mid-payment	Checking your payment…	Resume status by idempotency key on reconnect; never resend a charge
Limit exceeded	This payment is above your limit. Verify your identity to raise it.	Offer KYC (FR-ONB-06)
No final state in 2 min	Your payment is being checked. We'll notify you within 30 minutes.	PENDING_REVIEW; alert ops

7. Architecture, integrations, data and APIs

XenDirekt owns the app, the back end and the admin console; one licensed partner handles card acceptance, FX and the MoMo payout.

XenDirekt orchestrates; the licensed partner moves the money



system architecture · 9 components

The app talks only to the XenDirekt back end; the back end is the only component that talks to the partner, and the partner is the only one that touches card networks and mobile operators.

Recommended stack (developer may propose alternatives with reasons)

Layer	Recommendation
Mobile app	Flutter or React Native (one codebase for iOS and Android)
Back end	Node.js (NestJS) or Kotlin (Spring Boot), REST + JSON, OpenAPI spec
Database	PostgreSQL (payments, users, ledger), Redis (quotes, rate limits, idempotency keys)
Admin console	React web app behind SSO + 2FA
Hosting	Managed cloud with an African region; data location to be confirmed against Rwanda's data protection law (Law n° 058/2021)
Messaging	SMS gateway with delivery reports to +250 numbers and international numbers; FCM/APNs for push; transactional email
Monitoring	Error tracking (e.g. Sentry), structured logs, uptime alerts to the on-call phone

Partner integration requirements. The chosen partner must offer, through API: (1) card tokenisation and 3-D Secure, (2) separate authorise, capture and void, (3) MoMo account-name lookup for MTN and Airtel, (4) MoMo credit to a phone number and to a MoMo Pay merchant code, (5) webhooks for every

status change, (6) a sandbox, and (7) a daily settlement file. Candidates to evaluate: Flutterwave, DPO Pay, Pesapal, TerraPay or Thunes for payout, and MTN MoMo's own developer API.

Core data model

Entity	Key fields
User	id, name, email, phone (E.164), country, date of birth, PIN hash, KYC level, status, created_at
Card	id, user_id, partner_token, brand, last4, expiry, issuing country, fingerprint, is_default, status
Payee	id, network (MTN/AIRTEL), type (PHONE/MERCHANT_CODE), identifier, registered_name, last_used_at
Quote	id, user_id, payee_id, amount_rwf, fee_rwf, fx_rate, card_currency, total_card_amount, expires_at
Payment	id, reference (XD-XXXXXX), user_id, card_id, payee_id, quote_id, idempotency_key, state, partner_auth_id, partner_payout_id, failure_code, timestamps per state
PaymentEvent	id, payment_id, from_state, to_state, source (app/partner/ops), raw_payload (masked), created_at
Ticket	id, user_id, payment_id, category, message, status, assignee
AuditLog	id, actor, action, entity, before, after, created_at

All money is stored as integers in minor units with a currency code; RWF has no decimals.

App ↔ back end API (v1)

Method & path	Purpose
POST /v1/auth/otp, POST /v1/auth/verify	Send and verify the one-time code; returns access + refresh tokens
GET/PATCH /v1/me	Read and update the profile
POST /v1/cards, GET /v1/cards, DELETE /v1/cards/{id}	Add (with partner token), list, remove cards
POST /v1/payees/lookup	Body: network, type, identifier → registered name or error
POST /v1/quotes	Body: payee_id, amount_rwf, card_id → fee, FX rate, total, expiry
POST /v1/payments	Header Idempotency-Key; body: quote_id, note → payment with state and 3-D Secure URL if needed

Method & path	Purpose
GET /v1/payments, GET /v1/payments/{id}	History and live status
POST /v1/payments/{id}/tickets	Report a problem on a payment
POST /v1/webhooks/partner	Partner status callbacks; signature verified, idempotent

8. Non-functional requirements

Fraud and data security are the highest-risk areas: a stolen card paid into a MoMo account that can be cashed out at once is the main loss scenario, so the risk controls below are Must for the pilot.

8.1 Security and privacy

ID	Requirement
NFR-SEC-01	No card number, CVV or full PAN ever reaches XenDirekt servers or logs; only partner tokens (keeps XenDirekt out of full PCI DSS scope, SAQ-A level)
NFR-SEC-02	TLS 1.2+ everywhere; certificate pinning in the app
NFR-SEC-03	Data at rest encrypted (AES-256); secrets in a managed vault, never in code
NFR-SEC-04	Access tokens expire after 15 minutes; refresh tokens bound to the device; PIN or biometrics to open the app after 5 minutes in background
NFR-SEC-05	Phone numbers, names and payloads masked in logs and in the admin console for Viewer role
NFR-SEC-06	Partner webhooks verified by signature and source IP; replayed events ignored
NFR-SEC-07	Root/jailbreak detection and screenshot blocking on payment screens
NFR-SEC-08	Comply with Rwanda's personal data protection law (Law n° 058/2021): consent, purpose limitation, right to deletion, registration with the regulator if required

8.2 Fraud and risk controls

ID	Requirement
NFR-RSK-01	3-D Secure on every first payment per card, and whenever the partner's risk score asks for it
NFR-RSK-02	Limits per user, per card and per payee (amount per payment, per day, per trip), configurable in admin; starting values set by the PM with the partner

ID	Requirement
NFR-RSK-03	Velocity rules: many cards on one device, one card on many accounts, many tourists paying one new payee in a short time → flag for review
NFR-RSK-04	Block cards whose name or country does not match the user profile when the partner returns that data
NFR-RSK-05	Device fingerprinting; one active account per device
NFR-RSK-06	Sanctions and PEP screening of users through the partner
NFR-RSK-07	Every flagged payment is held at AUTHORISED and sent to the review queue before payout

8.3 Performance, availability and quality

ID	Requirement
NFR-PER-01	Payee lookup returns in ≤ 3 s (p95); quote in ≤ 1 s (p95)
NFR-PER-02	XenDirekt API adds ≤ 300 ms (p95) on top of partner response time
NFR-PER-03	App usable on a 3G connection; payment screens under 200 KB of data per payment
NFR-AVL-01	Back end availability 99.5% monthly during the pilot; partner outages shown in-app with a clear banner
NFR-AVL-02	Daily database backups, 30-day retention, tested restore
NFR-QA-01	Automated tests for every state transition in section 6, including duplicate webhooks and timeouts
NFR-QA-02	Accessibility: text scales with system font size, contrast ratio ≥ 4.5:1, all buttons labelled for screen readers
NFR-QA-03	Supported devices: Android 9+ and iOS 15+

9. Brand and UI guidelines

The app uses a dark-first look: black surfaces, white text and the XenDirekt orange as the single accent for primary actions.

Logo. Two interlocking chevrons forming an X: the orange chevron points right (send), the black or white one points left (receive). Use the black version on light backgrounds and the white version on dark backgrounds. Keep clear space around the mark equal to half its width; never stretch, recolour or add effects.





Colour palette

Token	Hex	Use
brand-orange	#FF640A	Primary buttons, active tab, key amounts, progress
brand-black	#000000	App background (dark theme), logo on light backgrounds
brand-white	#FFFFFF	Text on dark, light-theme background, logo on dark
surface-1	#1E1E1E	Cards, sheets and input fields on the black background
text-secondary	#A0A0A0	Helper text and labels on dark (proposed)
status-success	#22A559	Payment completed (proposed)
status-error	#E5484D	Errors, declined, failed (proposed)

Contrast rule. Orange on black passes for text (about 7:1). Orange on white does not (about 3:1), so orange buttons carry **black** text, and orange is never used for body text on white.

Typography. One sans-serif family for the whole app (Inter or the platform default). Amounts are large and bold (32–40 pt) on the confirmation and success screens.

Screens to design (MVP)

1. Splash (logo centred on black, as in the mock-up above) and one welcome screen with sign-in buttons
2. Sign up, OTP, profile, PIN and biometrics
3. Add card (partner hosted fields) and card list
4. Home: the **Pay** screen is the home screen (the app opens on it), with recent payees as one-tap chips and the last 3 payments
5. Pay: network toggle (MTN / Airtel), number or merchant-code field, amount keypad in RWF
6. Live summary on the Pay screen (no separate confirm screen): merchant name in large type, amount, fee, total in card currency, "Check the name with the seller" hint
7. Processing (live status steps) → Success (orange check, receipt) or Failure (reason + next step)
8. History list and receipt detail with share
9. Help centre and report a problem
10. Settings: language, cards, security, delete account

Tone of copy. Short, plain and reassuring: always say whether the user was charged. Example: "Payment failed. You have not been charged."

10. Analytics, risks, open questions and delivery

10.1 Analytics events

Event	Key properties
app_installed, signup_started, signup_completed	country, language, device
card_added, card_add_failed	brand, issuing country, failure reason
payee_lookup_succeeded / failed	network, type (phone/code), latency ms
quote_shown, payment_confirmed	amount_rwf, card currency, fee
payment_completed / failed / voided	state, failure code, time to pay (s), settlement time (s)
ticket_created	category, payment state

No personal data (names, numbers) in analytics; use internal IDs only.

10.2 Risks

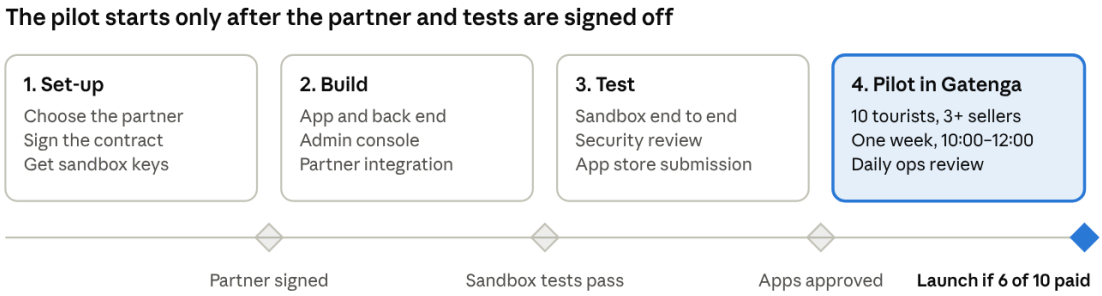
Risk	Impact	Mitigation
Card fraud: stolen cards paid into MoMo and cashed out	High	Section 8.2 controls, low starting limits, review queue
Issuers treat payments as money transfers or cash advances, adding fees for the tourist	High	Agree the right merchant category with the partner; test with cards from 5+ countries in the pilot

Risk	Impact	Mitigation
Partner cannot pay MoMo Pay merchant codes or return account names	High	Make both a must-have in partner selection
SMS to foreign numbers not delivered	Medium	Fall back to push + email; show receipt in-app
Tourists don't trust paying a stranger	High	Name shown before paying, clear receipts; this is the pilot's main test
Licensing delay	Medium	Launch only under the partner's licence; start BNR discussions early

10.3 Open questions

- ☐Which licensed partner? Confirm the 7 capabilities in section 7 in writing.
- ☐Fee model: flat fee, percentage, or both? Who pays FX spread?
- ☐Starting limits per payment, per day and per trip.
- ☐Is KYC (passport + selfie) needed from the first payment, or only above a limit?
- ☐Where may personal data be hosted under Rwanda's data protection law?
- ☐Merchant SMS: sent by XenDirekt, by the partner, or both? Sender name "XenDirekt" registered?
- ☐Card-network category for these payments and its effect on tourist fees.

10.4 Delivery roadmap



delivery roadmap · 4 phases, 4 gates

Each phase starts only when the gate before it is met; durations are set with the developer after estimation.

10.5 MVP acceptance criteria

1. A tester with a foreign Visa or Mastercard pays a real MTN and a real Airtel number, and a MoMo Pay code, in the partner's production environment.
2. The registered name appears before every payment; an unknown number blocks payment.
3. Merchant and tourist both receive confirmation within 30 seconds of success.
4. A forced payout failure releases the card hold and the tourist sees "You have not been charged."

5. Resending a payment with the same idempotency key never creates a second charge.
6. Admin can find any payment by reference, see its full timeline, refund it and export the daily reconciliation.
7. No card number appears in any XenDirekt log, database or analytics event (verified by a security review).

Sources

- XenDirekt Week 5 Deliverables and value proposition document (Bonae, Future Makers programme)
- National Bank of Rwanda — Regulation governing payment service providers (2023)
- TerraPay — MoMo Rwanda partnership for international transfers (Nov 2025)
- WorldRemit — Mobile money to Rwanda
- Bento Travel — Payments in Rwanda for visitors